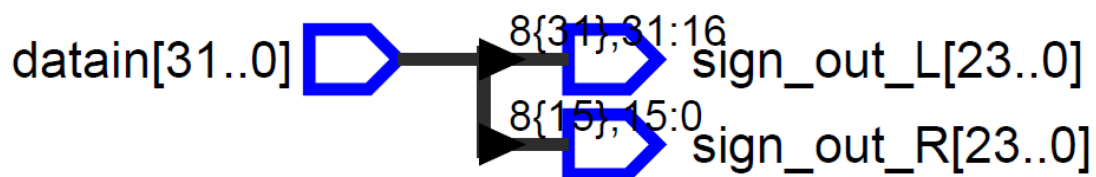


SignExtender Block:

This block will take in datain sample of 32 bits and split it into 2 16bits before performing a sign extension of 8 bits base on the MSB (rightmost bit of each half). The final outputs will be sign_out_L and sign_out_R each of which s 24 bits.

<https://edaplayground.com/x/Wm3Q>

Block diagram of SignExtender: shows datain[31:0] as input and sign_out_L/R [23:0]



TestBench output of the SignExtender Block it splits datain two L and R

```
# run -all
# Starting SignExtender Testbench...
# Time=0 | datain=0x4bc812a4 | sign_out_L=0x004bc8 | sign_out_R=0x0012a4
# Time=10000 | datain=0x910f368d | sign_out_L=0xff910f | sign_out_R=0x00368d
# Time=20000 | datain=0x1eddafb9 | sign_out_L=0x001edd | sign_out_R=0xffafb9
# Time=30000 | datain=0x80000000 | sign_out_L=0xff8000 | sign_out_R=0x000000
# Time=40000 | datain=0x7fff7fff | sign_out_L=0x007fff | sign_out_R=0x007fff
# Time=50000 | datain=0xffff0000 | sign_out_L=0xffffffff | sign_out_R=0x000000
# Time=60000 | datain=0x0000ffff | sign_out_L=0x000000 | sign_out_R=0xffffffff
# Time=70000 | datain=0x1234abcd | sign_out_L=0x001234 | sign_out_R=0xffabcd
# Testbench finished.
```